

Kamil Mikolaj

Machine Learning Engineer

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Summary

ML engineer with a track record of owning systems end-to-end across MLOps, computer vision, and 3D reconstruction. Comfortable at every layer of the stack — from C++ to Kubernetes-based retraining pipelines. Background in a domain where evaluation rigour and data quality aren't optional, which shapes how I approach production ML generally.

Experience

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| Dalux , Machine Learning and Computer Vision Engineer
Applying ML and computer vision to BIM and construction site understanding — building production systems across data pipelines, model training, and inference, with visibility into cloud deployment (AWS, Kubernetes, Argo) and monitoring (Prometheus/Grafana). | Copenhagen
Nov 2025 – present
7 months |
| GE Healthcare , Visiting Researcher
Research and optimization of motion-compensated 3D ultrasound reconstruction for laparoscopic liver workflows. <ul style="list-style-type: none">Targeted sub-minute runtime per sequence. | Herlev
Jan 2025 – Mar 2025
2 months |
| Technical University of Denmark (DTU) , PhD Student
End-to-end ML/CV research in obstetric ultrasound in collaboration with Rigshospitalet. <ul style="list-style-type: none">Technical lead for 3D reconstruction, fetal-growth modeling, heart segmentation, and analysis.Cut training time by 3x at matched quality for a 3D ultrasound reconstruction pipeline. PublicationBuilt an automated large-scale ultrasound pipeline for image-based fetal growth regression.Reduced measurement error by 13% versus current clinical practice. PublicationContributed to two UK patents. Granted, PendingBuilt a heart segmentation model with end-user input and deployed a real-time tablet prototype. | Kongens Lyngby
Aug 2022 – Oct 2025
3 years 3 months |
| SPS - Specialpaedagogisk Stotte , Pedagogical Support Teacher, Part-time
Providing academic and mentoring support to students with special educational needs. | Kongens Lyngby
Apr 2024 – Dec 2025
1 year 9 months |
| Copenhagen Academy for Medical Education and Simulation (CAMES) , Research Assistant, Part-time
Collaborative work between DTU and Rigshospitalet on ML in obstetric ultrasound. <ul style="list-style-type: none">Contributed to data preprocessing, fetal growth prediction, and multi-institutional project coordination. | Copenhagen
June 2021 – June 2022
1 year 1 month |

Education

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| PhD Technical University of Denmark (DTU) , Applied Mathematics and Computer Science (Visual Computing) <ul style="list-style-type: none">Thesis: From Automated 2D Analysis to 3D Reconstruction: Advancing Ultrasound Imaging with AI-Driven Methods. DownloadCourses | Kongens Lyngby
Aug 2022 – Oct 2025 |
| MSc Technical University of Denmark (DTU) , Autonomous Systems Engineering (Honours Programme) <ul style="list-style-type: none">Thesis: Automatic Detection of Anatomical Structures in Fetal Hearts.Courses | Kongens Lyngby
Sept 2020 – June 2022 |

Skills

Modelling: Deep Learning, Segmentation, 3D Reconstruction, NeRF, Semi-supervised Learning, Unsupervised Learning, Classification

Engineering: Python, C/C++, Systems Design, Inference Optimization, API Design

Infrastructure: MLOps, Cloud Deployment, Workflow Orchestration, Monitoring, Containerization

Data: Pipeline Design, Annotation, Dataset Curation, Data Quality Assurance

Computer Vision: Image Processing, Image Analysis, 3D Reconstruction

Tools

ML Frameworks: PyTorch, OpenCV, scikit-learn

Infrastructure & Cloud: AWS, Kubernetes, Docker, Argo Workflows, Argo Rollouts

Monitoring & Observability: Prometheus, Grafana, Loki

MLOps: MLflow, ONNX, Helm

Languages: Python, C/C++, MATLAB, LaTeX

Awards

UK Patent Application GB2636226

2023

A method of, and apparatus for, improved estimation of fetal characteristics.

UK Patent Application GB2636227

2023

An improved method of, and apparatus for, ultrasound examination to extract fetal characteristics.